

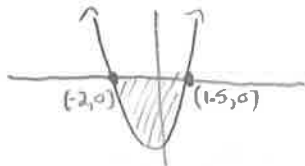
e) $2x^2 + x - 6 < 0$

~~$\frac{-1 \pm \sqrt{1+48}}{4}$~~

$(x+2)(2x-3) < 0$

x-int at $x = -2, 1.5$

+ L.C. } Q2 → Q1
Even Degree



Solution:

when $-2 < x < 1.5$

OR

when $x \in (-2, 1.5)$

f) $x^3 - 2x^2 - 13x \leq 10$

$f(-1) = 0$

or $x+1$ is a factor

$x^3 - 2x^2 - 13x - 10 \leq 0$

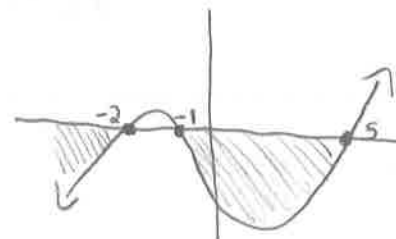
-1	1	-2	-13	-10
	1	-1	3	10
x	1	-3	-10	0
	x^2	x	#	R

$(x+1)(x^2 - 3x - 10) \leq 0$

$(x+1)(x-5)(x+2) \leq 0$

x-int at $x = -2, -1, 5$

+ L.C. } Q3 → Q1
odd degree



Solution:

when $x \leq -2$ or $-1 \leq x \leq 5$

when $x \in (-\infty, -2] \cup [-1, 5]$

Unit 3: Exponential and Logarithmic Functions

18) Rewrite each equation in logarithmic form.

a) $5^4 = 625$

$\log_5(625) = 4$

b) $4^x = 12$

$\log_4(12) = x$

c) $y = 12^3$

$\log_{12}(y) = 3$

19) Rewrite each equation in exponential form.

a) $x = \log 8$

$10^x = 8$

b) $4 = \log_5 x$

$5^4 = x$

c) $7 = \log_b 200$

$b^7 = 200$

20) Evaluate without using a calculator.

a) $\log_2 256$

$= \log_2(2^8)$

$= 8$

b) $\log_4 64$

$= \log_4(4^3)$

$= 3$

c) $\log_2 0.25$

$= \log_2(2^{-2})$

$= -2$

$$d) 2^{\log 1000}$$

$$= 2^3$$

$$= 8$$

$$e) \log\left(\frac{1}{1000}\right)$$

$$= \log(10)^{-3}$$

$$= -3$$

$$f) \log_6 6$$

$$= 1$$

21) Solve each exponential equation. Use and show appropriate methods. Round to 3 decimal places where necessary. Make sure to check for extraneous routes where necessary.

$$a) 3^x = 12$$

$$\log_3(12) = x$$

$$\frac{\log(12)}{\log(3)} = x$$

$$x \approx 2.26$$

$$b) 4^{2x+5} = 32^{4-x}$$

$$(2^2)^{2x+5} = (2^5)^{4-x}$$

$$2^{4x+10} = 2^{20-5x}$$

$$4x+10 = 20-5x$$

$$9x = 10$$

$$x = \frac{10}{9}$$

$$c) 27^{2-3x} = \left(\frac{1}{9}\right)^{2x}$$

$$(3^3)^{2-3x} = (3^{-2})^{2x}$$

$$3^{6-9x} = 3^{-4x}$$

$$6-9x = -4x$$

$$6 = 5x$$

$$x = \frac{6}{5}$$

$$d) 4^{2x-5} = 7^x$$

$$\log(4)^{2x-5} = \log(7)^x$$

$$(2x-5)\log(4) = x\log(7)$$

$$2x\log(4) - 5\log(4) = x\log(7)$$

$$2x\log(4) - x\log(7) = 5\log(4)$$

$$x(2\log(4) - \log(7)) = 5\log(4)$$

$$x = \frac{5\log(4)}{2\log(4) - \log(7)}$$

$$x \approx 8.385$$

$$e) 5^{x+4} = 2^{4-5x}$$

$$\log(5)^{x+4} = \log(2)^{4-5x}$$

$$(x+4)\log(5) = (4-5x)\log(2)$$

$$x\log(5) + 4\log(5) = 4\log(2) - 5x\log(2)$$

$$x\log(5) + 5x\log(2) = \log(16) - \log(625)$$

$$x(\log(5) + 5\log(2)) = \log(16) - \log(625)$$

$$x = \frac{\log(16) - \log(625)}{\log(5) + 5\log(2)}$$

$$x \approx -0.722$$

$$f) 3^{2x} - 4(3)^x + 3 = 0$$

$$\text{Let } k = 3^x$$

$$k^2 - 4k + 3 = 0$$

$$(k-3)(k-1) = 0$$

$$k=3 \text{ or } k=1$$

Case 1:

$$3^x = 3$$

$$x = 1$$

Case 2:

$$3^x = 1$$

$$\log_3(1) = x$$

$$\frac{\log(1)}{\log(3)} = x$$

$$x = 0$$

22) Write as a single logarithm and then evaluate. Round to 2 decimal places if necessary.

a) $\log_4 12 - \log_4 3$

$$= \log_4 \left(\frac{12}{3} \right)$$

$$= \log_4 (4)$$

$$= 1$$

b) $3 \log 6 + 2 \log 5 - \log 54$

$$= \log(6)^3 + \log(5)^2 - \log(54)$$

$$= \log(216) + \log(25) - \log(54)$$

$$= \log \left(\frac{216 \times 25}{54} \right)$$

$$= \log 100$$

$$= 2$$

23) Simplify.

a) $\log(x^2 - 4x - 12) - \log(3x - 18)$

$$= \log \left(\frac{x^2 - 4x - 12}{3x - 18} \right)$$

$$= \log \left[\frac{(x-6)(x+2)}{3(x-6)} \right]$$

$$= \log \left(\frac{x+2}{3} \right)$$

b) $\log(x^3 - 27) - \log(x - 3)$

$$= \log \left(\frac{x^3 - 27}{x - 3} \right)$$

$$= \log \left[\frac{(x-3)(x^2 + 3x + 9)}{x-3} \right]$$

$$= \log(x^2 + 3x + 9)$$

24) Solve the following logarithmic equations. Use and show appropriate methods. Round to 3 decimal places where necessary. Make sure to check for extraneous roots where necessary.

a) $\log_3(3x + 7) = 2$

$$3^2 = 3x + 7$$

$$9 = 3x + 7$$

$$2 = 3x$$

$$\boxed{x = \frac{2}{3}}$$

b) $\log_5(2x + 1) = 1 - \log_5(x + 2)$

$$\log_5(2x + 1) + \log_5(x + 2) = 1$$

$$\log_5[(2x + 1)(x + 2)] = 1$$

$$5^1 = 2x^2 + 5x + 2$$

$$0 = 2x^2 + 5x - 3$$

$$0 = 2x^2 - 1x + 6x - 3$$

$$0 = x(2x - 1) + 3(2x - 1)$$

$$0 = (2x - 1)(x + 3)$$

d) $\log_4(x^2 - 9x + 18) - \log_4(x - 3) = 2$

$$\log_4 \left(\frac{x^2 - 9x + 18}{x - 3} \right) = 2$$

$$\log_4 \left[\frac{(x-3)(x-6)}{x-3} \right] = 2$$

$$\log_4(x - 6) = 2$$

$$4^2 = x - 6$$

$$16 = x - 6$$

$$\boxed{x = 22}$$

c) $\log_4 x = \log_4 15 - \log_4 3$

$$\log_4(x) = \log_4 \left(\frac{15}{3} \right)$$

$$\log_4(x) = \log_4(5)$$

$$\boxed{x = 5}$$

$\boxed{x_1 = \frac{1}{2}}$ $\boxed{x_2 = -3}$
extraneous root

Exponential Formulas		
$A(t) = A_0(1 + i)^t$ where i is percent growth (+) or decay (-)	$A(t) = A_0 \left(\frac{1}{2}\right)^{\frac{t}{H}}$ where H is the half-life period	$A(t) = A_0(2)^{\frac{t}{D}}$ where D is the doubling period
Logarithmic Formulas		
$pH = -\log[H^+]$ where pH is acidity and $[H^+]$ is concentration of hydronium ions in mol/L	$\beta_2 - \beta_1 = 10 \log \left(\frac{I_2}{I_1}\right)$ where β is the loudness in dB and I is the intensity of sound in W/m^2	$M = \log \left(\frac{I}{I_0}\right)$ Where M is the magnitude measured by richters and I is intensity.

25) If the pH of a solution is 4.2, what is the concentration of the hydronium ions?

$$\begin{aligned}
 4.2 &= -\log[H^+] \\
 -4.2 &= \log[H^+] \\
 10^{-4.2} &= [H^+] \\
 [H^+] &\approx 0.0000630957 \quad \text{OR} \quad 6.31 \times 10^{-5}
 \end{aligned}$$

26) An investment earns interest compounded annually for 12 years. In that time, its value grows from \$2500 to \$7100. What was the interest rate, to the nearest tenth of a percent?

$$\begin{aligned}
 7100 &= 2500(1+i)^{12} \\
 2.84 &= (1+i)^{12} \\
 (2.84)^{1/12} &= 1+i \\
 (2.84)^{1/12} - 1 &= i
 \end{aligned}$$

$$i \approx 0.091$$

∴ The interest rate was about 9.1 %

27) A 30-mg sample of a radioactive isotope decays to 27 mg in 12.5 h.

a) Calculate its half-life, to two decimal places.

$$\begin{aligned}
 27 &= 30 \left(\frac{1}{2}\right)^{12.5/H} \\
 0.9 &= \left(\frac{1}{2}\right)^{12.5/H} \\
 \log_{0.5}(0.9) &= \frac{12.5}{H}
 \end{aligned}$$

$$H \approx 82.24 \text{ hours}$$

$$H = \frac{12.5}{\log_{0.5}(0.9)}$$

b) How long will it take (to the nearest hour) until only 5 mg of the sample remain?

$$\begin{aligned}
 5 &= 30 \left(\frac{1}{2}\right)^{t/82.24} \\
 \frac{1}{6} &= \left(\frac{1}{2}\right)^{t/82.24} \\
 \log_{0.5}\left(\frac{1}{6}\right) &= \frac{t}{82.24}
 \end{aligned}$$

$$t = 82.24 \log_{0.5}\left(\frac{1}{6}\right)$$

$$t \approx 212.59 \text{ hours}$$

28) The population of a species of animal in a nature reserve grows by 12.2% each year. Initially, there are 200 of that species.

a) Write an equation for the population of the species as a function of time, in years.

$$P(t) = 200(1.122)^t$$

b) What will the population be after 20 years?

$$P(20) = 200(1.122)^{20}$$

$$P(20) \approx 1999$$

c) How long does it take the population to double?

$$400 = 200(1.122)^t$$

$$2 = 1.122^t$$

$$\log 2 = \log 1.122^t$$

$$\log 2 = t \log 1.122$$

$$t = \frac{\log 2}{\log 1.122}$$

$$t \approx 6.02 \text{ years}$$

29) a) The intensity of a sound at the threshold of hearing (0 dB) is 10^{-12} W/m^2 . What is the intensity of a 50 dB sound?

β_2

$$50 - 0 = 10 \log \left(\frac{I_2}{10^{-12}} \right)$$

$$5 = \log \left(\frac{I_2}{10^{-12}} \right)$$

$$10^5 = \frac{I_2}{10^{-12}}$$

$$I_2 = (10^5)(10^{-12})$$

$$I_2 = 10^{-7}$$

$$I_2 = 10^{-7} \text{ W/m}^2$$

$$\beta_2 - \beta_1 = 10 \log \left(\frac{I_2}{I_1} \right)$$

b) How many times as intense is an 85 dB sound than a 50 dB sound?

β_2

β_1

$$85 - 50 = 10 \log \left(\frac{I_2}{I_1} \right)$$

$$3.5 = \log \left(\frac{I_2}{I_1} \right)$$

$$10^{3.5} = \frac{I_2}{I_1}$$

$$\frac{I_2}{I_1} \approx 3162.28$$

About 3162 times as intense

c) A noise is 400 times as intense as a 60 dB sound. What is the decibel rating of this noise?

$\frac{I_2}{I_1}$

β_1

$$\beta_2 - 60 = 10 \log (400)$$

$$\beta_2 = 10 \log (400) + 60$$

$$\beta_2 \approx 86.02 \text{ dB}$$

$$M = \log\left(\frac{I}{I_0}\right)$$

30)a) The magnitudes of two earthquakes are 4.7 and 7.1. How many times as intense was the stronger earthquake than the less severe one?

$$7.1 - 4.7 = \log\left(\frac{I}{I_0}\right)$$

$$10^{2.4} = \frac{I}{I_0}$$

$$\frac{I}{I_0} \approx 251.2$$

About 251.2 times
as intense

b) An earthquake is detected that is 450 times as intense as an earthquake with a magnitude of 5.2. What is the magnitude of the new earthquake?

$$M - 5.2 = \log(450)$$

$$M = \log(450) + 5.2$$

$$M \approx 7.85$$

Unit 4: Trig in Radians

31) Determine the approximate degree measure, to the nearest tenth, for each angle.

$$3.62 \times \frac{180}{\pi}$$

$$\approx 207.4^\circ$$

$$b) 0.54 \times \frac{180}{\pi}$$

$$\approx 30.9^\circ$$

32) Determine the exact radian measure of each angle.

$$a) 125^\circ \times \frac{\pi}{180}$$

$$= \frac{25\pi}{36} \text{ radians}$$

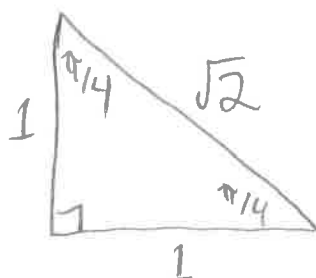
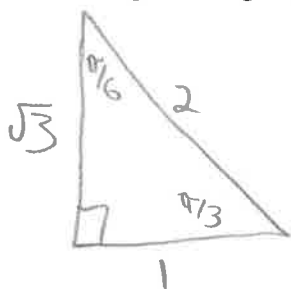
$$b) 80^\circ \times \frac{\pi}{180}$$

$$= \frac{4\pi}{9} \text{ radians}$$

$$c) 16^\circ \times \frac{\pi}{180}$$

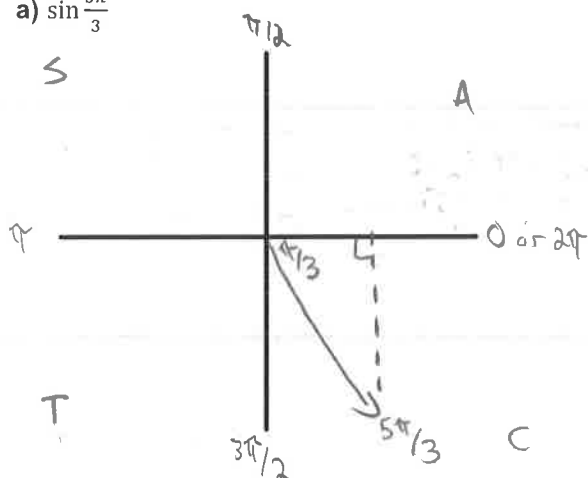
$$= \frac{4\pi}{45} \text{ radians}$$

33) Draw both special triangles using radian measures.



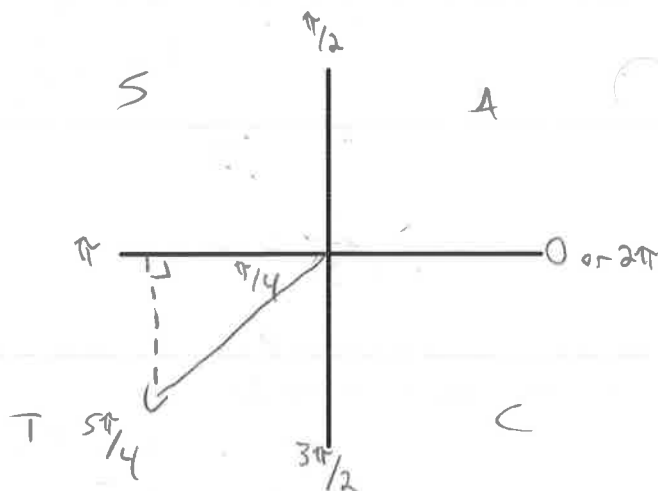
34) Find the exact value of the expressions below. Use special triangles and the CAST rule to find your answer.

a) $\sin \frac{5\pi}{3}$



$$\sin\left(\frac{5\pi}{3}\right) = -\sin\left(\frac{\pi}{3}\right) = -\frac{\sqrt{3}}{2}$$

b) $\cot \frac{5\pi}{4}$



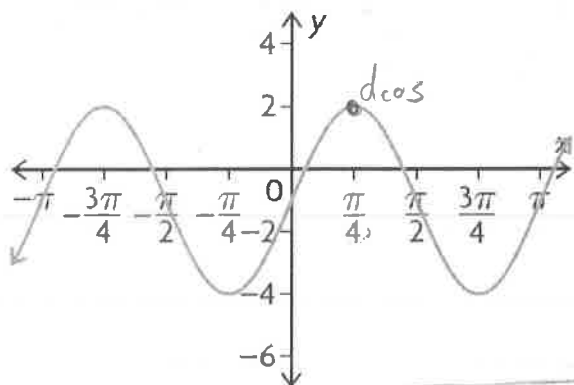
$$\cot\left(\frac{5\pi}{4}\right) = \frac{1}{\tan\left(\frac{5\pi}{4}\right)} = \frac{1}{\tan\left(\frac{\pi}{4}\right)} = \frac{1}{1} = 1$$

35) Determine an exact value for each expression.

a) $\frac{\sin \frac{\pi}{6} \tan \frac{\pi}{3}}{\csc \frac{\pi}{4}} = \frac{\left(\frac{1}{2}\right)\left(\frac{\sqrt{3}}{1}\right)}{\left(\frac{\sqrt{2}}{1}\right)} = \frac{\sqrt{3}}{2} \left(\frac{1}{\sqrt{2}}\right) = \frac{\sqrt{3}}{2\sqrt{2}} \left(\frac{\sqrt{2}}{\sqrt{2}}\right) = \frac{\sqrt{6}}{4}$

b) $\sec \frac{4\pi}{3} \cot \frac{5\pi}{6} - \tan \frac{3\pi}{4} = \frac{1}{\cos\left(\frac{4\pi}{3}\right)} \times \frac{1}{\tan\left(\frac{5\pi}{6}\right)} - \tan\left(\frac{3\pi}{4}\right) = \frac{1}{-\cos\left(\frac{\pi}{3}\right)} \times \frac{1}{-\tan\left(\frac{\pi}{6}\right)} - (-\tan \frac{\pi}{4}) = -2 \times -\sqrt{3} + 1 = 2\sqrt{3} + 1$

36) Find two equations (one sine and one cosine) to represent the function on the graph below. Show your calculations for full marks.



$$a = \frac{\max - \min}{2} = \frac{2 - (-4)}{2} = 3$$

$$K = \frac{2\pi}{\text{period}} = \frac{2\pi}{\pi} = 2$$

$$c = \max - |a| = 2 - 3 = -1$$

$$d_{\cos} = \frac{\pi}{4}$$

$$d_{\sin} = d_{\cos} - \frac{\pi}{2K} = \frac{\pi}{4} - \frac{\pi}{2(2)} = 0$$

$$y = 3 \cos\left[2\left(x - \frac{\pi}{4}\right)\right] - 1$$

$$y = 3 \sin(2x) - 1$$

37) A sine function has a maximum value of 6, a minimum value of -2, a period of $\frac{\pi}{2}$, and a phase shift of $\frac{\pi}{6}$ radians to the right.

Write an equation for the function

$$a = \frac{6 - (-2)}{2} = 4 \quad d \sin = \frac{\pi}{6}$$

$$k = \frac{2\pi}{(\frac{\pi}{2})} = 4$$

$$c = \max - |a| = 6 - 4 = 2$$

$$y = 4 \sin \left[4 \left(x - \frac{\pi}{6} \right) \right] + 2$$

b) Write an equivalent cosine equation for the function

$$d \cos = d \sin + \frac{\pi}{2k} = \frac{\pi}{6} + \frac{\pi}{2(4)} = \frac{4\pi}{24} + \frac{3\pi}{24} = \frac{7\pi}{24}$$

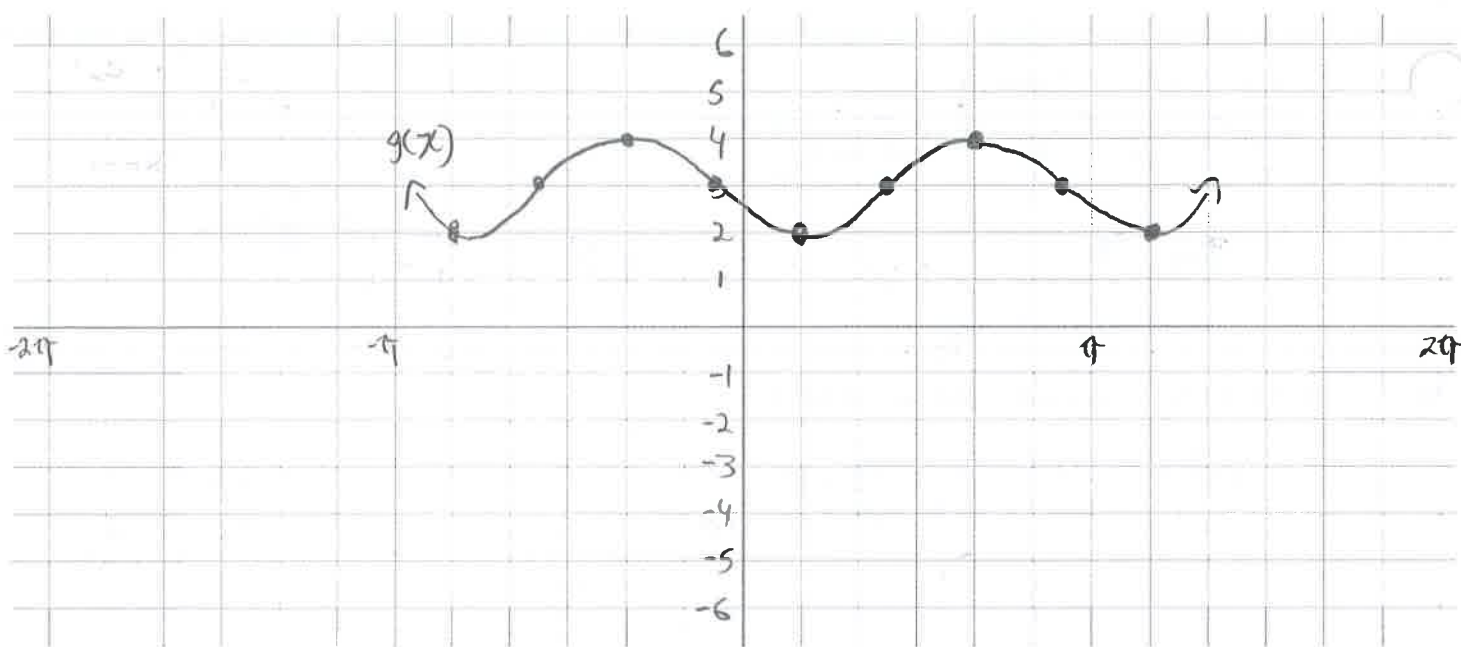
$$y = 4 \cos \left[4 \left(x - \frac{7\pi}{24} \right) \right] + 2$$

38) For the function $g(x) = -\cos \left[2 \left(x - \frac{\pi}{6} \right) \right] + 3$, fill in the table of information and then graph two cycles of the transformed function using transformations of the parent function. Choose an appropriate scale. [7]

Amplitude: $= a = 1$	Period: $= \frac{2\pi}{k} = \frac{2\pi}{2} = \pi$ radians
Phase shift: $\frac{\pi}{6}$ RIGHT	Vertical shift: 3 UP
Max: $= c + a = 3 + 1 = 4$	Min: $= c - a = 3 - 1 = 2$

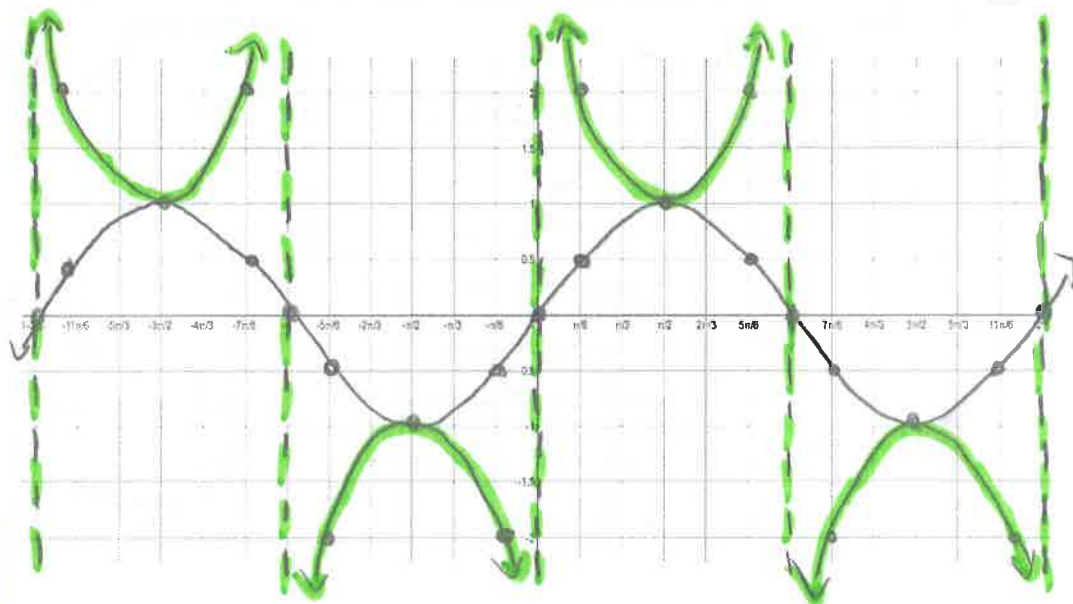
$y = \cos x$	
x	y
0	1
$\frac{\pi}{2}$	0
π	-1
$\frac{3\pi}{2}$	0
2π	1

$g(x)$	
$\frac{x}{2} + \frac{\pi}{6}$	$-y + 3$
$\frac{\pi}{6}$	2
$\frac{5\pi}{12} = \frac{2.5\pi}{6}$	3
$\frac{4\pi}{6}$	4
$\frac{11\pi}{12} = \frac{5.5\pi}{6}$	3
$\frac{7\pi}{6}$	2



39) Complete the following table of values for the function $f(x) = \sin(x)$ and $g(x) = \csc(x)$. Use special triangles, the unit circle, or a calculator to find values for the function. Then graph both functions on the same grid. Draw asymptotes where necessary.

x	$f(x)$	$g(x)$
0	0	und
$\frac{\pi}{6}$	$\frac{1}{2}$	2
$\frac{2\pi}{6} = \frac{\pi}{3}$	$\frac{\sqrt{3}}{2} = 0.87$	$\frac{2}{\sqrt{3}} = 1.15$
$\frac{3\pi}{6} = \frac{\pi}{2}$	1	1
$\frac{4\pi}{6} = \frac{2\pi}{3}$	0.87	1.15
$\frac{5\pi}{6}$	$\frac{1}{2}$	2
$\frac{6\pi}{6} = \pi$	0	und
$\frac{7\pi}{6}$	$-\frac{1}{2}$	-2
$\frac{8\pi}{6} = \frac{4\pi}{3}$	-0.87	-1.15
$\frac{9\pi}{6} = \frac{3\pi}{2}$	-1	-1
$\frac{10\pi}{6} = \frac{5\pi}{3}$	-0.87	-1.15
$\frac{11\pi}{6}$	$-\frac{1}{2}$	-2
$\frac{12\pi}{6} = 2\pi$	0	und



40) A Ferris wheel at an amusement park completes one revolution every 60 seconds. The wheel has a radius of 20 meters and its center is 22 meters above the ground. Assume the rider starts at the bottom.

a) Model the rider's height above the ground with a sine function

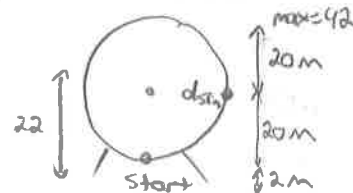
$$a = \frac{42 - 2}{2} = 20$$

$$d \sin = \frac{60}{4} = 15$$

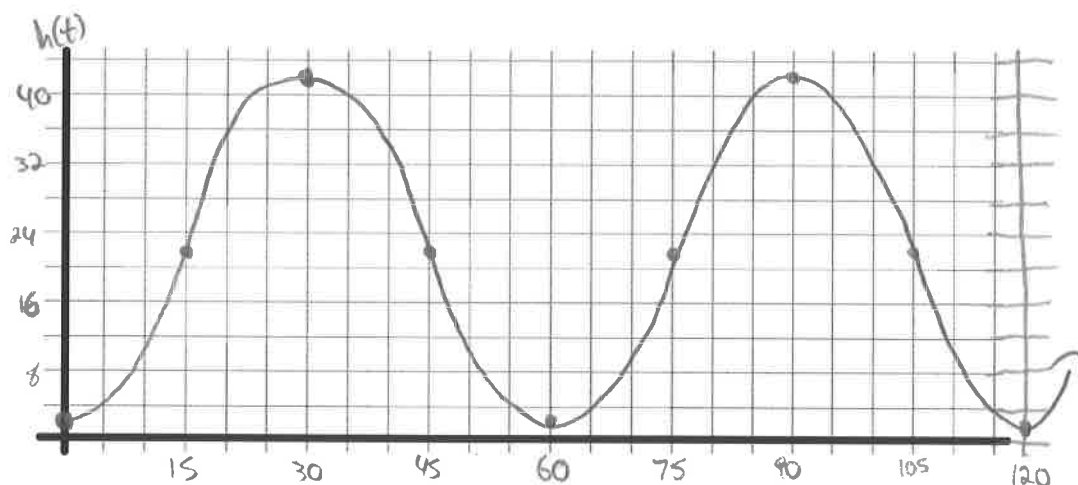
$$k = \frac{2\pi}{60} = \frac{\pi}{30}$$

$$c = 42 - 20 = 22$$

$$h(t) = 20 \sin \left[\frac{\pi}{30}(t - 15) \right] + 22$$



b) Sketch a graph of the rider's height above the ground for 2 cycles.



Unit 5: Trig Identities and Equations

41) Use an appropriate compound angle formula to express as a single trig function, and then determine an exact value for each.

a) $\sin \pi \cos \frac{\pi}{2} + \cos \pi \sin \frac{\pi}{2}$

$$= \sin \left(\pi + \frac{\pi}{2} \right)$$

$$= \sin \left(\frac{3\pi}{2} \right)$$

$$= -1$$

b) $\cos \pi \cos \frac{\pi}{2} + \sin \pi \sin \frac{\pi}{2}$

$$= \cos \left(\pi - \frac{\pi}{2} \right)$$

$$= \cos \left(\frac{\pi}{2} \right)$$

$$= 0$$

42) Use an appropriate compound angle formula to determine an exact value for each.

a) $\cos \frac{\pi}{12}$

$$= \cos \left(\frac{4\pi}{12} - \frac{3\pi}{12} \right)$$

$$= \cos \left(\frac{\pi}{3} - \frac{\pi}{4} \right)$$

$$= \cos \frac{\pi}{3} \cos \frac{\pi}{4} + \sin \frac{\pi}{3} \sin \frac{\pi}{4}$$

$$= \left(\frac{1}{2} \right) \left(\frac{1}{\sqrt{2}} \right) + \left(\frac{\sqrt{3}}{2} \right) \left(\frac{1}{\sqrt{2}} \right)$$

$$= \frac{1 + \sqrt{3}}{2\sqrt{2}}$$



b) $\sin \frac{11\pi}{12}$

$$= \sin \left(\frac{8\pi}{12} + \frac{3\pi}{12} \right)$$

$$= \sin \left(\frac{2\pi}{3} + \frac{\pi}{4} \right)$$

$$= \sin \frac{2\pi}{3} \cos \frac{\pi}{4} + \cos \frac{2\pi}{3} \sin \frac{\pi}{4}$$

$$= \sin \frac{\pi}{3} \cos \frac{\pi}{4} + (-\cos \frac{\pi}{3}) \left(\sin \frac{\pi}{4} \right)$$

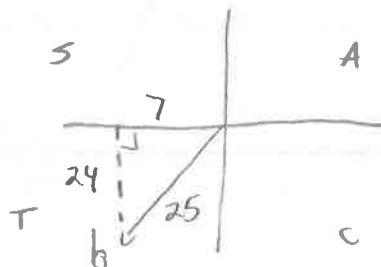
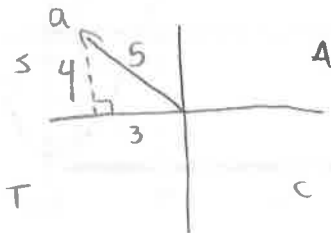
$$= \frac{\sqrt{3}}{2} \left(\frac{1}{\sqrt{2}} \right) - \frac{1}{2} \left(\frac{1}{\sqrt{2}} \right)$$

$$= \frac{\sqrt{3} - 1}{2\sqrt{2}}$$

43) Angle a lies in the second quadrant and angle b lies in the third quadrant such that $\cos a = -\frac{3}{5}$ and $\tan b = \frac{24}{7}$. Determine an exact value for

2,214297436

diagrams:



a) $\cos(a + b)$

$$= \cos a \cos b - \sin a \sin b$$

$$= \left(-\frac{3}{5} \right) \left(-\frac{7}{25} \right) - \left(\frac{4}{5} \right) \left(-\frac{24}{25} \right)$$

$$= \frac{21 + 96}{125}$$

$$= \frac{117}{125}$$

b) $\sin(a - b)$

$$= \sin a \cos b - \cos a \sin b$$

$$= \left(\frac{4}{5} \right) \left(-\frac{7}{25} \right) - \left(-\frac{3}{5} \right) \left(-\frac{24}{25} \right)$$

$$= \frac{-28 - 72}{125}$$

$$= \frac{-100}{125}$$

$$= -\frac{4}{5}$$

c) $\sin(2a)$

$$= 2 \sin a \cos a$$

$$= 2 \left(\frac{4}{5} \right) \left(-\frac{3}{5} \right)$$

$$= -\frac{24}{25}$$

d) $\cos(2b)$

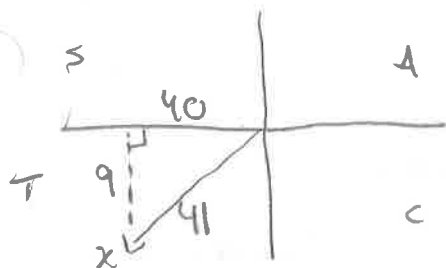
$$= \cos^2 b - \sin^2 b$$

$$= \left(-\frac{7}{25} \right)^2 - \left(-\frac{24}{25} \right)^2$$

$$= \frac{49 - 576}{625}$$

$$= -\frac{527}{625}$$

44) Angle x lies in the third quadrant, and $\tan x = \frac{9}{40}$. Determine an exact value for $\cos(2x)$.



$$\begin{aligned}\cos(2x) &= \cos^2 x - \sin^2 x \\ &= \left(-\frac{40}{41}\right)^2 - \left(-\frac{9}{41}\right)^2 \\ &= \frac{1600 - 81}{1681} \\ &= \frac{1519}{1681}\end{aligned}$$

45) Prove the following identities. Use a separate piece of paper.

a) $\sec x - \tan x = \frac{1 - \sin x}{\cos x}$

b) $(\csc x - \cot x)^2 = \frac{1 - \cos x}{1 + \cos x}$

c) $\sin(2a) = \frac{2 \tan a}{\sec^2 a}$

d) $\cos(x + y) \cos(x - y) = \cos^2 x + \cos^2 y - 1$

e) $\frac{\cos(2x)}{1 - \sin(2x)} = \frac{\cos x + \sin x}{\cos x - \sin x}$

f) $\sin(2x) = 2 \sin x \cos x$

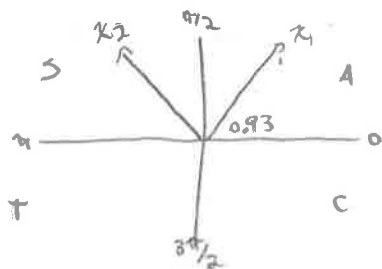
g) $\frac{\tan x - \tan y}{\cot x - \cot y} = -\tan x \tan y$

h) $\frac{2 \tan x}{1 + \tan^2 x} = \sin(2x)$

) Determine solutions for each equation in the interval $0 \leq x \leq 2\pi$, to the nearest hundredth of a radian. Give exact answers where possible.

a) $\sin x - 0.8 = 0$

$$\sin x = 0.8$$



$$x_1 = \sin^{-1}(0.8)$$

$$x_1 \approx 0.93$$

$$x_2 = \pi - 0.93$$

$$x_2 \approx 2.21$$

b) $\tan x - \frac{3}{4} = 0$

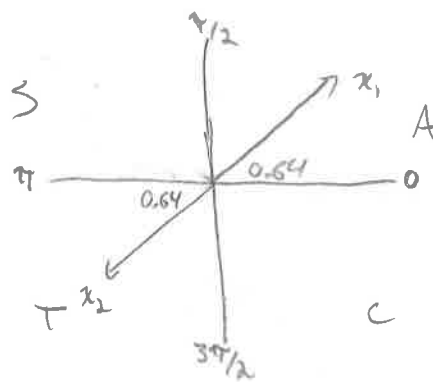
$$\tan x = \frac{3}{4}$$

$$x_1 = \tan^{-1}\left(\frac{3}{4}\right)$$

$$x_1 \approx 0.64$$

$$x_2 = \pi + 0.64$$

$$x_2 \approx 3.78$$



c) $2 \sin x = -\sqrt{3}$
 $\sin x = -\frac{\sqrt{3}}{2}$

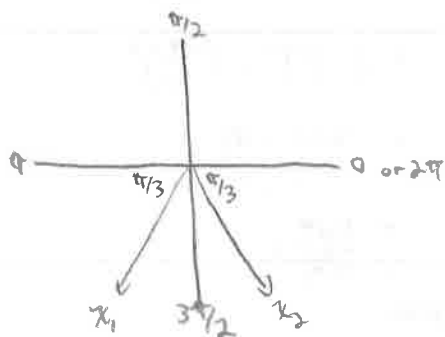
From Δ ; $\sin \frac{\pi}{3} = \frac{\sqrt{3}}{2}$
 place in Q3+Q4

$x_1 = \pi + \frac{\pi}{3}$

$x_1 = \frac{4\pi}{3}$

$x_2 = 2\pi - \frac{\pi}{3}$

$x_2 = \frac{5\pi}{3}$



d) $2 \sin x \cos x - \cos x = 0$

$\cos x (2 \sin x - 1) = 0$

✓
 $\cos x = 0$
 From unit circle:

$x_1 = \frac{\pi}{2}$

$x_2 = \frac{3\pi}{2}$

$2 \sin x - 1 = 0$

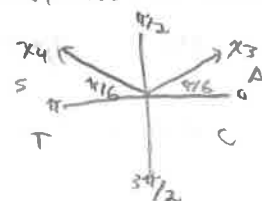
$\sin x = \frac{1}{2}$

From Δ ; $\sin \frac{\pi}{6} = \frac{1}{2}$

place in Q1+Q2

$x_3 = \frac{\pi}{6}$

$x_4 = \frac{5\pi}{6}$



e) $\csc^2 x = 2 + \csc x$

$\csc^2 x - \csc x - 2 = 0$

$(\csc x - 2)(\csc x + 1) = 0$

✓
 $\csc x = 2$
 $\sin x = \frac{1}{2}$

✓
 $\csc x = -1$
 $\sin x = -1$

from part d)

$x_1 = \frac{\pi}{6}$

$x_2 = \frac{5\pi}{6}$

from unit circle:

$x_3 = \frac{3\pi}{2}$

g) $64 \sin^2 x - 25 = 0$

$\sin^2 x = \frac{25}{64}$

$\sin x = \pm \sqrt{\frac{25}{64}}$

$\sin x = \pm \frac{5}{8}$

✓
 $\sin x = \frac{5}{8}$

$x_1 \approx 0.675$

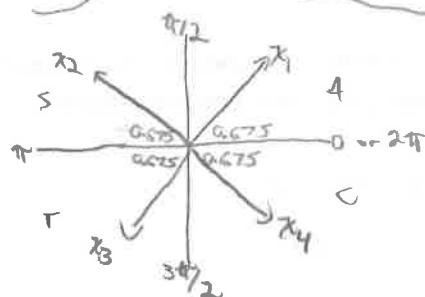
$x_2 \approx 2.47$

$\sin x = -\frac{5}{8}$

place 0.675 in Q3+Q4

$x_3 \approx 3.82$

$x_4 \approx 5.61$

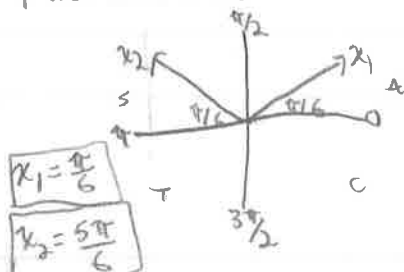


h) $2 \csc^2 x - 9 \csc x + 10 = 0$

$(\csc x - 2)(2 \csc x - 5) = 0$

✓
 $\csc x = 2$
 $\sin x = \frac{1}{2}$

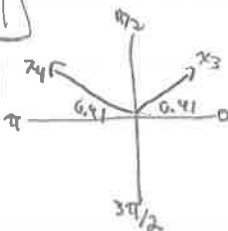
From Δ ; $\sin \frac{\pi}{6} = \frac{1}{2}$
 place in Q1+Q2



$x_3 = \sin^{-1}(\frac{2}{5})$

$x_3 \approx 0.41$

$x_4 \approx 2.73$



i) $2 \cos(2x) = 1$

$\cos(2x) = \frac{1}{2}$ let $\theta = 2x$

$\cos \theta = \frac{1}{2}$

From Δ ; $\cos \frac{\pi}{3} = \frac{1}{2}$

place in Q1+Q4

$\theta_1 = \frac{\pi}{3}$ $\theta_2 = \frac{5\pi}{3}$

✓
 $2x = \frac{\pi}{3}$

$x_1 = \frac{\pi}{6}$

$x_3 = x_1 + \pi$

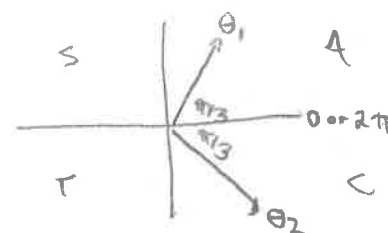
$x_3 = \frac{7\pi}{6}$

✓
 $2x = \frac{5\pi}{3}$

$x_2 = \frac{5\pi}{6}$

$x_4 = x_2 + \pi$

$x_4 = \frac{11\pi}{6}$



* Add period of π
 to find other answers

Unit 6: Rational Equations/Inequalities and Rates of Change

47) Solve each equation

a) $\frac{x-2}{3x-1} = \frac{2}{x+1}$

$$(x+1)(x-2) = 2(3x-1)$$

$$x^2 - x - 2 = 6x - 2$$

$$x^2 - 7x = 0$$

$$x(x-7) = 0$$

$$x_1 = 0$$

$$x-7=0$$

$$x_2 = 7$$

b) $3 = \frac{6}{2x^2 - x - 4}$

$$3(2x^2 - x - 4) = 6$$

$$6x^2 - 3x - 12 = 6$$

$$6x^2 - 3x - 18 = 0$$

$$3(2x^2 - x - 6) = 0$$

$$(2x^2 - 4x + 3x - 6) = 0$$

$$[2x(x-2) + 3(x-2)] = 0$$

$$(x-2)(2x+3) = 0$$

$$x-2=0$$

$$2x+3=0$$

$$x_1 = 2$$

$$x_2 = -\frac{3}{2}$$

48) Solve each inequality

a) $\frac{2x-3}{x+5} > \frac{2x+7}{x-3}$

$$\frac{2x-3}{x+5} - \frac{2x+7}{x-3} > 0$$

$$\frac{(x-3)(2x-3) - (2x+7)(x+5)}{(x+5)(x-3)} > 0$$

$$\frac{2x^2 - 3x - 6x + 9 - (2x^2 + 10x + 7x + 35)}{(x+5)(x-3)} > 0$$

$$\frac{2x^2 - 9x + 9 - 2x^2 - 17x - 35}{(x+5)(x-3)} > 0$$

$$\frac{-26x - 26}{(x+5)(x-3)} > 0$$

$$\frac{-26(x+1)}{(x+5)(x-3)} > 0$$

$$x\text{-int: } x = -1$$

$$\text{restrictions: } x \neq -5, 3$$

	$-\infty$	-5	-1	3	5	∞
$x+1$	-	-	+	+	+	+
$x+5$	-	+	+	+	+	+
$x-3$	-	-	-	+	+	+
overall	+	-	+	-	+	+

Solution:

$$x < -5 \text{ or } -1 < x < 3$$

$$x \in (-\infty, -5) \cup (-1, 3)$$

b) $\frac{x^2 - 8x + 15}{x^2 + 5x + 4} \leq 0$

$$\frac{(x-3)(x-5)}{(x+4)(x+1)} \leq 0$$

$$x\text{-ints: } x = 3, 5$$

$$\text{restrictions: } x \neq -4, -1$$

	$-\infty$	-4	-1	3	5	∞
$x-3$	-	-	-	+	+	+
$x-5$	-	-	-	-	+	+
$x+4$	-	+	+	+	+	+
$x+1$	-	-	+	+	+	+
overall	+	-	+	-	+	+

Solution:

$$-4 < x < -1 \text{ or } 3 \leq x \leq 5$$

$$x \in (-4, -1) \cup [3, 5]$$

49) The population of a small town, p , is modelled by the function $p(t) = 10\,050 + 225t - 20t^2$, where t is the time in years from now. Determine the average rate of change of the population from

a) year 0 to year 5

$$\begin{aligned}
 m &= \frac{\Delta y}{\Delta x} \\
 &= \frac{p(5) - p(0)}{5 - 0} \\
 &= \frac{10\,675 - 10\,050}{5} \\
 &= 125 \text{ ppl/year}
 \end{aligned}$$

b) year 5 to year 8

$$\begin{aligned}
 m &= \frac{p(8) - p(5)}{8 - 5} \\
 &= \frac{10\,570 - 10\,678}{3} \\
 &= -36 \text{ ppl/year}
 \end{aligned}$$

50) A soccer ball is kicked into the air. The following table of values shows the height of the ball above the ground at various times during its flight:

Time (seconds)	0.0	0.5	1.0	1.5	2.0	2.5	3.0	3.5	4.0	4.5	5.0
Height (meters)	0.5	11.78	20.6	26.98	30.9	32.38	31.4	27.98	22.1	13.78	3.0

a) Find the average rate of change in the height of the ball over the first 3 seconds.

$$\begin{aligned}
 m &= \frac{h(3) - h(0)}{3 - 0} \\
 &= \frac{31.4 - 0.5}{3} \\
 &= 10.3 \text{ m/sec}
 \end{aligned}$$

b) Estimate the *instantaneous* rate of change of the height at 4 seconds.

Method 1: surrounding

$$\begin{aligned}
 \frac{dh}{dt} \Big|_{t=4} &\approx \frac{h(4.5) - h(3.5)}{4.5 - 3.5} \\
 &= \frac{13.78 - 27.98}{1} \\
 &= -14.2 \text{ m/sec}
 \end{aligned}$$

Method 2: Average preceding and following

$$\begin{aligned}
 m \text{ for } 3.5 \leq t \leq 4 \\
 m &= \frac{22.1 - 27.98}{4 - 3.5} \\
 &= -11.76
 \end{aligned}$$

$$\begin{aligned}
 m \text{ for } 4 \leq t \leq 4.5 \\
 m &= \frac{13.78 - 22.1}{4.5 - 4} \\
 &= -16.64
 \end{aligned}$$

$$\begin{aligned}
 \frac{dh}{dt} \Big|_{t=4} &\approx \frac{-11.76 + (-16.64)}{2} \\
 &= -14.2 \text{ m/sec}
 \end{aligned}$$

51) Use the chart below to calculate several average rates of change to help you estimate the instantaneous rate of change for the function $f(x) = -2x^2 + x$ at $x = 1$. Have 4-decimal place accuracy in the $\frac{\Delta y}{\Delta x}$ column.

Interval	Change in $y = \Delta y$	Δx	$\frac{\Delta y}{\Delta x} = \text{Avg. Rate of Change}$
$0 \leq x \leq 1$	$f(1) - f(0)$ $= -1 - 0$ $= -1$	$1 - 0$ $= 1$	$= \frac{-1}{1}$ $= -1$
$0.5 \leq x \leq 1$	$f(1) - f(0.5)$ $= -1 - 0$ $= -1$	$1 - 0.5$ $= 0.5$	$= \frac{-1}{0.5}$ $= -2$
$0.9 \leq x \leq 1$	$f(1) - f(0.9)$ $= -1 + 0.72$ $= -0.28$	$1 - 0.9$ $= 0.1$	$= \frac{-0.28}{0.1}$ $= -2.8$
$0.99 \leq x \leq 1$	$f(1) - f(0.99)$ $= -1 + 0.9702$ $= -0.0298$	$1 - 0.99$ $= 0.01$	$= \frac{-0.0298}{0.01}$ $= -2.98$

$$\left. \frac{dy}{dx} \right|_{x=1} \approx -3$$

Use the Newton Quotient to find the equation of the derivative for each of the following functions. Also, find the instantaneous rate of change for the function when $x = 4$.

a) $f(x) = x^2 + 6$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{(x+h)^2 + 6 - (x^2 + 6)}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 + 6 - x^2 - 6}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{2xh + h^2}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{h(2x+h)}{h}$$

$$f'(x) = 2x + 0$$

$$f'(x) = 2x$$

$$f'(4) = 2(4)$$

$$f'(4) = 8$$

b) $f(x) = 2x^2 + 3x - 4$

$$f'(x) = \lim_{h \rightarrow 0} \frac{2(x+h)^2 + 3(x+h) - 4 - (2x^2 + 3x - 4)}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{2(x^2 + 2xh + h^2) + 3x + 3h - 4 - 2x^2 - 3x + 4}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{2x^2 + 4xh + 2h^2 + 3x + 3h - 4 - 2x^2 - 3x + 4}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{4xh + 2h^2 + 3h}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{h(4x + 2h + 3)}{h}$$

$$f'(x) = 4x + 2(0) + 3$$

$$f'(x) = 4x + 3$$

$$f'(4) = 4(4) + 3$$

$$f'(4) = 19$$

53) Determine the equation of the tangent line at $x = 3$ for the function $f(x) = 5x^2 - 10x - 7$

$$f'(x) = \lim_{h \rightarrow 0} \frac{5(x+h)^2 - 10(x+h) - 7 - (5x^2 - 10x - 7)}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{5x^2 + 10xh + 5h^2 - 10x - 10h - 7 - 5x^2 + 10x + 7}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{h(10x + 5h - 10)}{h}$$

$$f'(x) = 10x + 5(0) - 10$$

$$f'(x) = 10x - 10$$

$$f'(3) = 10(3) - 10 = 20$$

∴ slope of tangent is 20.

$$f(3) = 5(3)^2 - 10(3) - 7 = 8$$

∴ (3, 8) is on tangent line

$$\begin{aligned} y &= mx + b \\ 8 &= 20(3) + b \\ b &= -52 \end{aligned}$$

$$y = 20x - 52$$

54) Calculate the following limits:

a) $\lim_{x \rightarrow 5} \frac{x^2 - 25}{x - 5}$

$$= \lim_{x \rightarrow 5} \frac{(x-5)(x+5)}{(x-5)}$$

$$= 5 + 5$$

$$= 10$$

b) $\lim_{x \rightarrow 3} \frac{2x^2 - 5x - 3}{x - 5}$

$$= \frac{2(3)^2 - 5(3) - 3}{3 - 5}$$

$$= \frac{0}{-2}$$

$$= 0$$

55) Calculate the following limits using the graph provided.

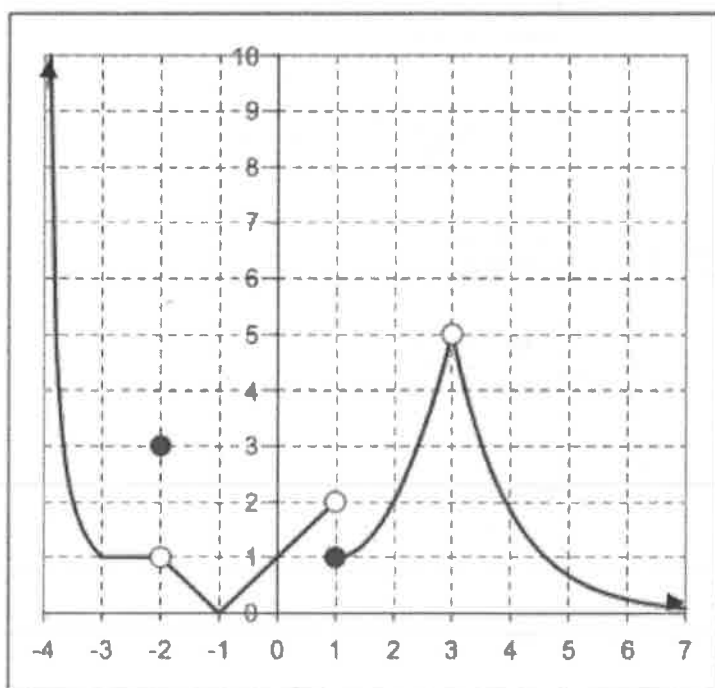
a) $\lim_{x \rightarrow \infty} f(x) = 0$

b) $\lim_{x \rightarrow -4^+} f(x) = \infty$

c) $\lim_{x \rightarrow 1^+} f(x) = 1$

d) $\lim_{x \rightarrow 1^-} f(x) = 2$

e) $\lim_{x \rightarrow 1} f(x)$ Does Not Exist



Unit 7: Graphing

56) Graph $f(x) = \frac{4}{x-2}$ by first graphing the denominator, then graphing its reciprocal. Don't forget about the
on top! Label any asymptotes

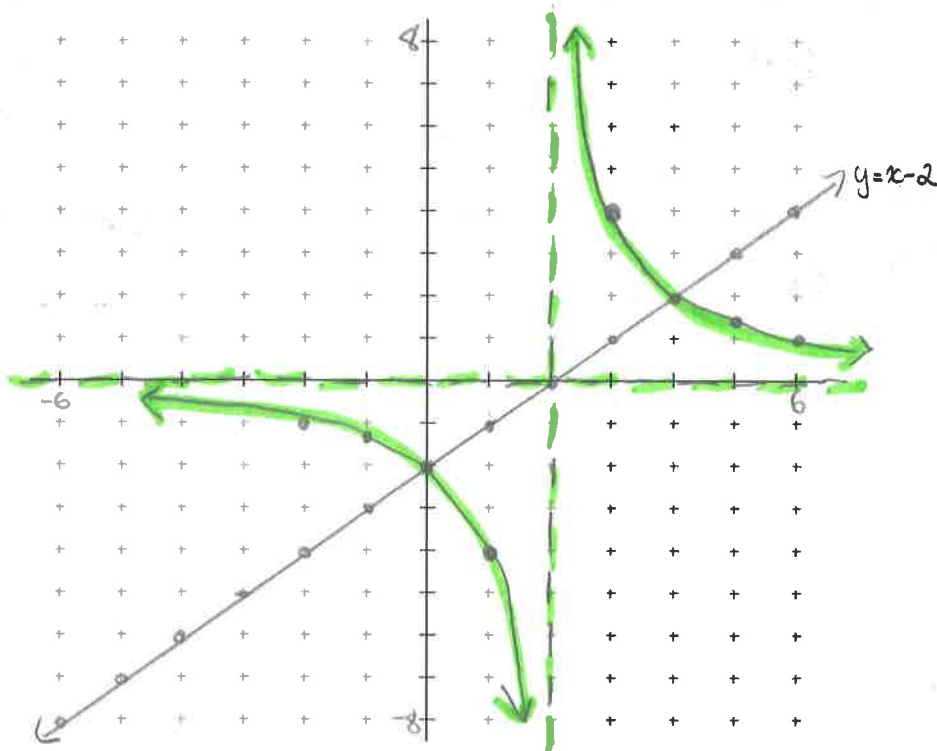
$$HA: y=0$$

$$VA: x=2$$

$$y = x - 2$$

x	y
-1	-3
0	-2
1	-1
2	0
3	1
4	2
5	3

$f(x) = \frac{4}{x-2}$	
x	$\frac{4}{y}$
-1	-1.33
0	-2
1	-4
2	undefined
3	4
4	2
5	1.33



57) Graph $g(x) = \frac{1}{x^2-4}$ by first graphing the denominator, then graph its reciprocal. Label any asymptotes.

$$g(x) = \frac{1}{(x-2)(x+2)}$$

$$HA: y=0$$

$$VA: x=-2 \text{ and } x=2$$

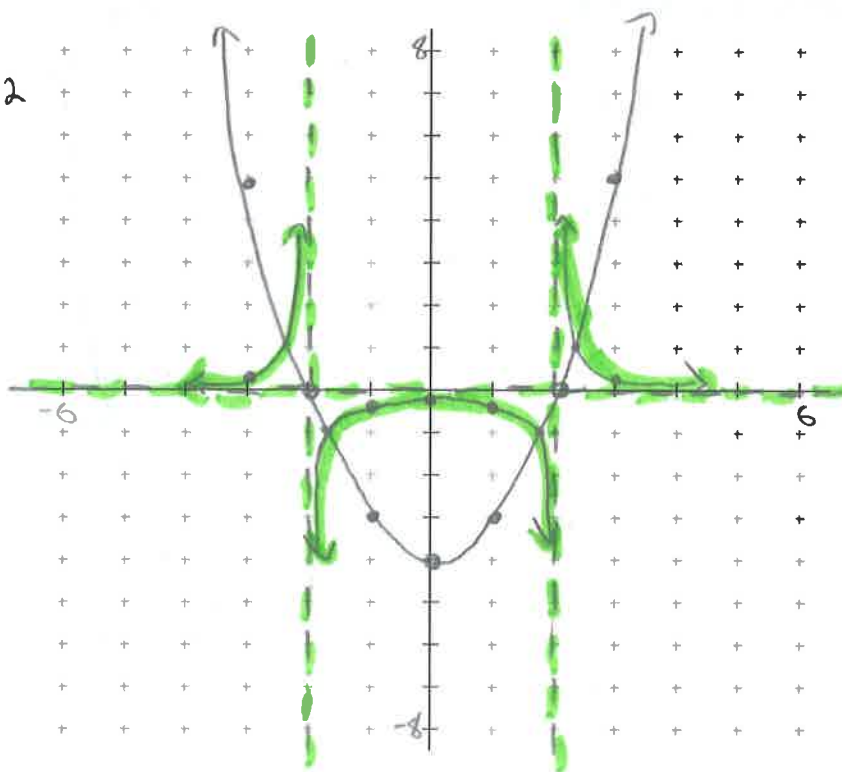
$$y = x^2 - 4$$

vertex at (0, -4)

x-inter at (-2, 0) and (2, 0)

x	y
-3	5
-2	0
-1	-3
0	-4
1	-3
2	0
3	5

$g(x) = \frac{1}{x^2-4}$	
x	$\frac{1}{y}$
-3	0.2
-2	und
-1	-0.33
0	-0.25
1	-0.33
2	und
3	0.2



58) Graph the function $h(x) = \frac{2x-3}{x+1}$. Clearly state the key features of this graph including asymptotes and intercepts.

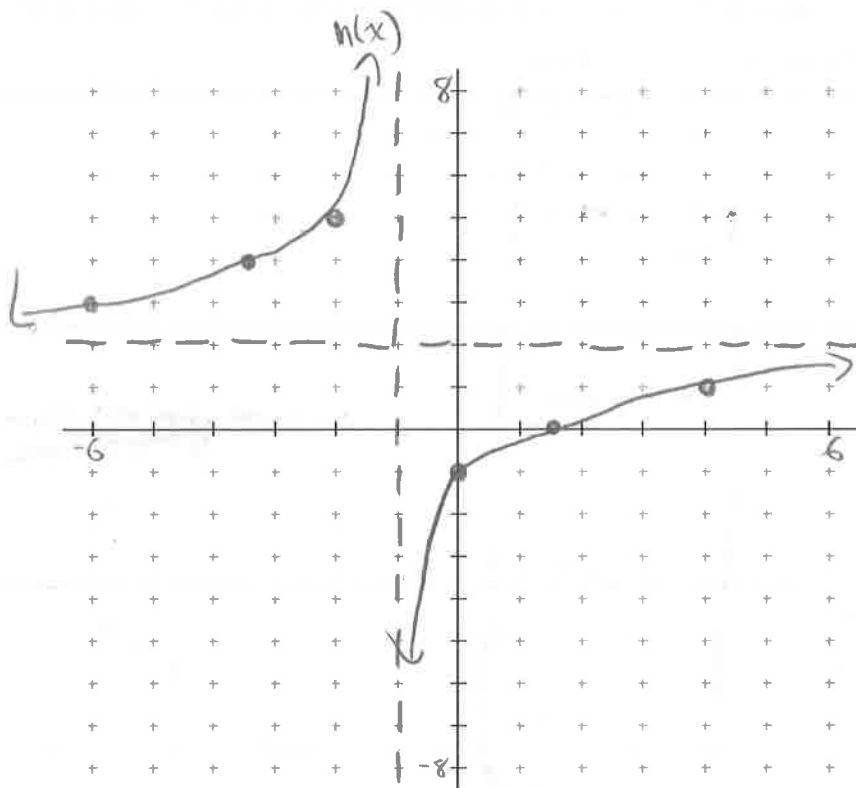
VA: $x = -1$
 HA: $y = 2$

x-int: $0 = 2x - 3$
 $x = 1.5$
 $(1.5, 0)$

y-int: $h(0) = \frac{-3}{1}$
 $(0, -3)$

other
point:

$h(4) = \frac{2(4)-3}{4+1}$
 $= 1$
 $(4, 1)$



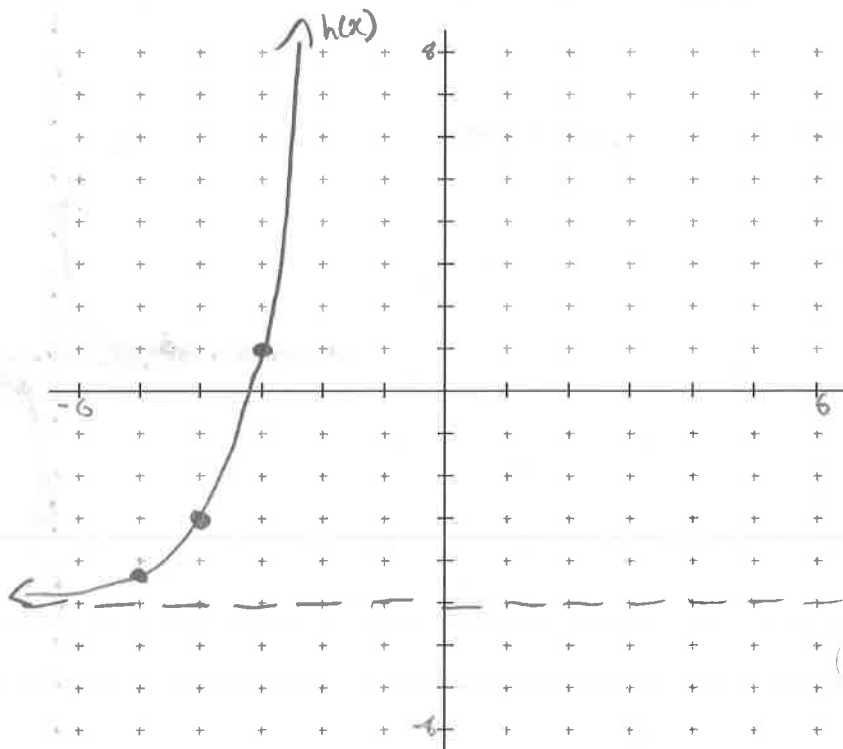
59) Graph the transformed function $h(x) = 2(3)^{x+4} - 5$ using transformations.

$y = 3^x$

x	y
-1	0.33
0	1
1	3
HA	$y = 0$

$h(x)$

$x-4$	$2y-5$
-5	-4.33
-4	-3
-3	-1
HA	$y = -5$



60) Evaluate each of the following, correct to 3 decimal places.

a) e^5

$$\approx 148.413$$

b) $\ln 200$

$$\approx 5.298$$

61) Simplify the following expression $\ln e^{2x}$

$$= 2x \ln(e)$$

$$= 2x (1)$$

$$= 2x$$

62) Solve the following equations, correct to 3 decimal places.

a) $e^x = 5$

$$x = \ln(5)$$

$$x \approx 1.609$$

b) $1000 = 20e^{\frac{x}{4}}$

$$50 = e^{x/4}$$

$$\frac{x}{4} = \ln(50)$$

$$x = 4 \ln(50) \approx 15.648$$

63) Given that $g(x) = x + 5$ and $f(x) = x^2 + 4x - 1$, find a formula for the following...

a) $f(g(x))$

$$= f(x+5)$$

$$= (x+5)^2 + 4(x+5) - 1$$

$$= x^2 + 10x + 25 + 4x + 20 - 1$$

$$= x^2 + 14x + 44$$

b) $g(f(x))$

$$= g(x^2 + 4x - 1)$$

$$= x^2 + 4x - 1 + 5$$

$$= x^2 + 4x + 4$$

c) $g^{-1}(f(x))$

$$\underline{g^{-1}(x)}$$

$$x = y + 5$$

$$y = x - 5$$

$$g^{-1}(x) = x - 5$$

$$\underline{g^{-1}(f(x))}$$

$$= g^{-1}(x^2 + 4x - 1)$$

$$= x^2 + 4x - 1 - 5$$

$$= x^2 + 4x - 6$$

d) $f(x) - g(x)$

$$= x^2 + 4x - 1 - (x + 5)$$

$$= x^2 + 4x - 1 - x - 5$$

$$= x^2 + 3x - 6$$

